REGIME-AWARE SIGNAL ENGINE · BTC/USDT · 1H

LATENT

A technical report on a regime-aware BTC/USDT research system: Kalman filtering for direction, a hidden-Markov model for market regime, GARCH for volatility, a stacked ensemble for the decision, and fractional Kelly for the size.

Author	Ganesh Tilekar
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Licence	MIT
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Status	Research. Backtested, not live traded.

Read this before anything else

This document describes a research tool. It is not investment advice, not financial advice, and not a recommendation to buy or sell any asset. No advisory relationship is created by reading it.

Every performance figure in this report comes from historical backtesting. There is no live track record. Hypothetical results are prepared with hindsight, do not represent actual trading, and cannot account for the effect of financial risk in live markets.

Trading cryptocurrency involves substantial risk of loss, including loss of your entire capital. You are solely responsible for your own decisions and for complying with the laws of your jurisdiction.

1 Introduction

Most technical indicators are functions of price that react to price. A moving average tells you where price has been. An oscillator tells you whether the recent move was large relative to other recent moves. Neither attempts to say anything about the *state* the market is currently in, and yet the same signal means very different things in a trending market and a choppy one.

LATENT is built around the opposite premise: estimate the state first, then decide. Three unobservable quantities are estimated at every bar — the underlying trend and its velocity, the prevailing market regime, and the volatility about to be experienced. Only then does a model decide whether to act, and a separate layer decides how much.

1.1 Objectives

- Build a noise-filtered, regime-aware signal pipeline.
- Engineer a dimensionless feature set from raw OHLCV alone.
- Train a stacked ensemble producing a calibrated probability per bar.
- Validate through walk-forward testing so no future information leaks.
- Size positions using fractional Kelly with hard risk caps.
- Report through institutional-standard metrics, including the unflattering ones.

1.2 Architecture

Five layers. Raw candles enter on the left, a sized order leaves on the right, and nothing downstream can see the future.

LAYER	COMPONENTS	RESPONSIBILITY
Data	load_data, add_base_features	Ingest hourly OHLCV, compute 30+ normalised columns
Signal	Kalman, HMM, Hurst, OFI, VWAP-z	Directional, regime and statistical state estimates
Model	XGBoost, LightGBM, meta-logistic	One calibrated probability per bar
Risk	Fractional Kelly, GARCH, drawdown brake	How much to commit, and when not to
Execution	6-condition gate, ATR TP/SL, timeout	Order placement, exit, realistic cost simulation

The separation is deliberate. If any single model fails to converge — the HMM, most commonly — the pipeline filters to whatever feature columns actually exist and continues in degraded mode. It would rather run on 25 features than crash on 29. That matters when real capital is involved.

2 Data pipeline

2.1 Dependencies

LIBRARY	ROLE
NumPy	Array mathematics throughout
Pandas	DataFrames, rolling windows, EWM

LIBRARY	ROLE
SciPy	norm.logpdf for HMM emissions; optimize.minimize (L-BFGS-B) for GARCH MLE
XGBoost	XGBClassifier — 600 trees, depth 5, learning rate 0.03
LightGBM	LGBMClassifier — 600 estimators, 40 leaves, learning rate 0.03
scikit-learn	LogisticRegression meta-learner, StandardScaler, roc_auc_score
Matplotlib	Equity curve and drawdown chart
openpyxl	Reading the Binance .xlsx export

The Kalman filter, the hidden-Markov model and the Hurst exponent are implemented directly rather than imported. This keeps the dependency list short enough to install anywhere, and means the mathematics is readable rather than hidden behind an API.

2.2 Loading and sorting

The loader accepts .xlsx or .csv, normalises column names against a table of common aliases, coerces types, drops unparseable rows, and then does the single most important thing in the entire codebase:

```
df = df.sort_values("date").drop_duplicates(subset="date", keep="last")
```

> **Why sorting is not a detail.** If even a handful of rows sit out of chronological order, every rolling window and the target column will silently incorporate future information into past rows. The backtest will look extraordinary and it will be fiction. This is the most common way a research result turns out to be worthless.

2.3 Validation

A separate `validate()` pass reports problems in plain language rather than failing silently: too few rows for the folds, gaps in the series, bars where high sits below low, open or close outside the high-low range, and an implausible share of zero-volume bars. Running with `--validate` performs these checks and exits.

3 Feature engineering

Wherever possible features are dimensionless. A feature that scales with price teaches the model that Bitcoin was cheap in 2020, which is true and useless. Dividing by ATR or by price makes the same information portable across price levels and across time.

GROUP	COLUMNS	NOTE
Returns	ret, log_ret	Log returns add over time, which makes multi-period analysis cleaner
Momentum	mom_3, mom_5, mom_10	Three horizons: short, medium, slightly longer
Trend	ema_9/21/50/200, fast_slow, trend_up	fast_slow normalises the EMA gap so it survives a doubling in price
Volatility	atr, atr_pct, realised_vol	atr_pct rather than raw ATR enters the model
Oscillators	rsi, rsi_smooth, macd, macd_hist	MACD divided by price; RSI smoothed over 5 bars to reduce jitter
Participation	vol_ratio, vol_trend	Distinguishes a genuine volume surge from a one-off spike
Candle shape	body_pct, upper_wick, lower_wick, range_pct	All divided by ATR. A long upper wick hints at selling pressure above

3.1 Target

The supervised target is binary: did the close rise over the following five bars? Formally `target = 1 if close[t+5] > close[t]`. The final five rows have no future to look at and are excluded from training rather than filled.

4 Signal layer

4.1 Kalman filter

A constant-velocity Kalman filter runs over the close series. The state is [level, velocity]; the filter observes level only and infers velocity. Each bar performs a predict step and an update step, weighting the new observation against the accumulated estimate by the Kalman gain.

```
predict: x_p = F x P_p = F P F' + Q update : y = z - H x_p S = H P_p H' + R K = P_p H' / S x = x_p + K y
```

Two parameters control behaviour. `process_var` encodes how much you believe the trend genuinely changes; raising it makes the filter quicker and noisier. `measure_var` encodes how noisy you believe observed price to be; raising it smooths harder. The output `kf_velocity` is divided by price so it remains comparable at any level.

4.2 Hidden Markov model

Three states with Gaussian emissions over log returns, fitted by Baum-Welch. The entire forward-backward pass runs in log space; on sequences of several thousand bars, probabilities in linear space underflow to zero within a few hundred steps.

Nothing tells the model what “bear” means. It finds three return distributions, and the states are relabelled afterwards in ascending order of mean return, which is why the labels are stable between runs. The transition matrix is initialised with 0.9 on the diagonal, encoding the prior that regimes persist rather than alternate bar to bar.

Fitting is capped at the most recent 8,000 observations. Baum-Welch is inherently sequential and its cost grows linearly with sequence length; prediction still runs over the full history.

4.3 Hurst exponent

Estimated by rescaled range over a rolling window. H above 0.5 indicates a trending series where momentum logic is appropriate; below 0.5 indicates mean reversion; around 0.5 is a random walk. Recomputed every five bars rather than every bar — over a 100-bar window the value barely moves between adjacent bars, and recomputing it each time costs substantial runtime for no additional information.

4.4 Microstructure proxies

True order-flow imbalance requires the order book, which OHLCV does not contain. The proxy used is where each bar closed within its own range, weighted by volume: closing near the high means buyers controlled that bar. `vwap_z` measures distance from rolling VWAP in standard deviations, clipped to ±5 so a single dislocation cannot dominate the feature.

5 Volatility modelling

Volatility clusters. Quiet hours follow quiet hours and violent ones follow violent ones. A rolling standard deviation reports what volatility *was*; GARCH forecasts what it is about to be, which is the quantity actually needed when deciding position size.

```
sigma_t^2 = omega + alpha * e_{t-1}^2 + beta * sigma_{t-1}^2
```

Parameters are estimated by maximum likelihood using L-BFGS-B under bounds that keep the process stationary (alpha + beta < 0.999) and the variance positive. If the optimiser fails to converge, the model falls back to sensible defaults rather than raising — the pipeline is designed to degrade, not die.

Persistence (alpha + beta) close to 1 means shocks decay slowly. Values in the region of 0.97 are normal for crypto and are not a sign of misfit. The derived `vol_regime` column is forecast volatility divided by its own expanding average: above 1 means conditions are hotter than this market's own normal.

6 Model layer

6.1 Base learners and stacking

XGBoost and LightGBM are trained on the same feature matrix. Both are gradient-boosted tree ensembles but they split differently — LightGBM grows leaf-wise and uses histogram binning — so their errors are usefully decorrelated. A logistic regression sits on top and learns when to trust which.

> **The meta-learner must not see its base models' training data.** If the stack is fitted on predictions the base models made about rows they were trained on, it learns their overfitting rather than their signal. Here the base models train on the first 75% of the training slice and the meta-learner is fitted on their predictions over the held-out remainder.

Every split is chronological. Shuffling a time series before splitting places future information in the training set and is the second most common way a backtest becomes meaningless.

6.2 Walk-forward validation

The usable data is divided into sequential folds. Fold k trains on everything preceding its test window and is scored only on that window. The model is refitted at each fold rather than carried forward, so each result reflects what would have been known at that time.

Train once and test once gives you a number. This gives you a range, and the range is the honest answer.

6.3 Interpreting AUC

On financial data an AUC of 0.60 is a strong result; 0.50 is a coin flip. The stack typically improves on either base model by 0.01 to 0.03, which sounds negligible until compounded over a thousand trades. If AUC sits at 0.50 the model has found nothing, and the correct response is to distrust the strategy rather than to lower the threshold until it trades.

7 Risk layer

7.1 Fractional Kelly

```
f* = (p * b - q) / b q = 1 - p, b = reward / risk
```

Kelly is growth-optimal *given a known edge*. The edge here is an estimate from a model with an AUC around 0.6, which is emphatically not known. Half Kelly is the standard compromise: it gives up roughly a quarter of the growth rate for a large reduction in the probability of a deep drawdown. Negative edge returns zero — declining to bet is a valid bet.

7.2 Caps and the drawdown brake

PARAMETER	DEFAULT	CONTROLS
`starting_capital`	100,000	Baseline for all return and drawdown figures
`max_risk_per_trade`	0.01	If the stop is hit, the loss cannot exceed 1% of equity
`max_drawdown_limit`	0.15	Emergency brake — halts new entries
`max_position_frac`	0.20	Concentration cap, even if Kelly asks for more
`signal_buy_thresh`	0.62	Confidence required before a long is considered
`signal_sell_thresh`	0.38	Probability at or below which a short is considered

The brake engages when drawdown passes the limit and releases only once equity recovers to within half of it. A system that keeps trading through its worst stretch is how a bad month becomes a blown account. Volatility scaling is applied before the caps: forecast volatility above its long-run average divides the size, so exposure falls before the storm rather than after it.

8 Execution

8.1 The entry gate

Six conditions must all pass before a position opens:

#	CONDITION	PURPOSE
1	probability beyond threshold	The model is actually confident
2	RSI between 30 and 70	Do not buy into an exhausted move
3	ATR finite and positive	Volatility is measurable, so stops can be placed
4	cooldown since last exit	Prevents re-entering the same noise
5	drawdown brake off	Risk layer has not halted trading
6	vol_regime below 3	Not in a volatility spike

8.2 Exits and cost model

Stop-loss and take-profit are both ATR-scaled, so they widen in volatile conditions and tighten in quiet ones without any parameter changing. A trending tape, indicated by Hurst above 0.55, earns a 15% wider target. Any position still open after 25 bars is closed at market — a thesis that has not worked within a day has usually stopped being a thesis.

The simulation is deliberately pessimistic. Commission and slippage are charged on both entry and exit. If both the stop and the target fall inside the same bar, the stop is assumed to have been hit first. Real fills are worse than modelled fills, and a backtest that assumes otherwise is telling you what you want to hear.

9 Metrics and reporting

METRIC	WHAT IT TELLS YOU
Sharpe	Return per unit of total risk, annualised for hourly bars
Sortino	The same, but upside surprises are not counted as risk
Calmar	Annualised return against the worst drawdown endured
Max drawdown	The deepest hole the equity curve digs
Profit factor	Gross wins over gross losses; below 1.3 is not viable
VaR 95%	The loss that should be exceeded on only 5% of trades
Longest losing run	How many consecutive losses to expect. Live will be worse
Time under water	Fraction of the period spent below a previous peak

Time under water and longest losing run are included deliberately. They are the numbers that determine whether a system can actually be held through a bad stretch, and they are almost never quoted in marketing material for precisely that reason.

Results are additionally split by the regime each trade was opened in. A system that only works in one regime is a system waiting for that regime to end, and the per-regime table is where that becomes visible.

9.1 Output files

FILE	CONTENTS
`metrics.csv`	Headline figures
`trades.csv`	Every trade: entry, exit, size, regime, exit reason, P&L
`regime_breakdown.csv`	Metrics split by bear / sideways / bull
`folds.csv`	Per-fold rows, AUCs, probability ranges, returns
`feature_importance.csv`	Averaged gain importance across both base models
`equity_curve.png`	Equity and drawdown

10 Results

> **This section is intentionally left for you to complete.**
>
> The numbers below are the design targets taken from the original system specification, not measured output. Fill them in from your own run against real market data, and delete this box once you have. Publishing target figures as though they were measured is the exact failure this report is written to avoid.

METRIC	DESIGN TARGET	YOUR RESULT
Net return	~ +20%	
Win rate	58 – 65%	
Sharpe	1.4 – 2.0	
Max drawdown	< 8%	
Profit factor	> 1.5	
Model AUC	~ 0.60	

Expected regime shape, again as a design target: bear regimes producing 15–25% of trades at 45–55% win rate; sideways 35–45% of trades at 55–65%; bull 35–45% at 60–70%. If a real run inverts that pattern entirely, suspect the data before crediting the model.

11 Limitations

- **No live track record.** Everything here is backtested.
- **Tuned for BTC/USDT 1H.** Output on other markets is untested.
- **Bear regimes are the weak point**, close to break-even.
- **The HMM does not always converge** on short datasets.
- **Backtests are optimistic by construction.** Real fills are worse.
- **Edges decay.** A model that tested well may stop working, and this is the normal life cycle of a strategy rather than a defect in it.

11.1 On repeated testing

It takes roughly twenty iterations of adjusting and re-testing on the same data to discover a false strategy at conventional significance levels. Walk-forward validation reduces this risk but does not remove it. Feature importance analysis is a better research tool than repeatedly re-running a backtest until it looks good.

Risk disclosure

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